

Tran Huy Thinh

Software Engineer Intern | Backend | C++

[redacted]

email redacted

 github.com/HuyThinhUET |  Hanoi, Vietnam

EDUCATION

VNU University of Engineering and Technology

Bachelor of Information Technology

Sep 2023 – Expected Jun 2027

Hanoi, Vietnam

- Cumulative GPA: 3.70 / 4.00
- Most recent academic-year GPA: 3.90 / 4.00

Thai Nguyen High School for the Gifted

Specialization: Informatics

Aug 2020 – May 2023

Thai Nguyen, Vietnam

- Overall GPA: 9.0 / 10; Informatics GPA: 9.8 / 10

EXPERIENCE

AI Research Assistant

Apr 2025 – Present

Intelligent Software Engineering Laboratory (iSE)

VNU-UET

- Multi-Agent Systems:** Designed and implemented components for automated multi-agent system (MAS) generation and error prediction in generated MAS code, combining ML-based features, prompt engineering, and LLM-based reasoning.
- Archival Document Processing:** Developed an AI-assisted pipeline combining rule-based checks, semantic search, clustering, ML/DL, and LLM APIs to organize, classify, and process large-scale digitized documents.

PUBLICATIONS

- Trung-Hieu Nguyen, **Huy-Thinh Tran**, Quang-Hung Trinh, Tung-Thuy Pham, Thu-Trang Nguyen, Son Nguyen, and Hieu Dinh Vo. *MASAC: Abstract-Then-Concretize Automated Design of Multi-Agent System*. In *Proceedings of the 17th International Conference on Knowledge and Systems Engineering (KSE 2025)*.
 - Co-developed the MASAC framework, contributing to system implementation, experimental design, and evaluation of LLM-based multi-agent system generation.

ACHIEVEMENTS

- Informatics Olympiad:** Third Prize - National Olympiad in Informatics (2023); Consolation Prize - National Olympiad in Informatics (2022); First Prize - Provincial Olympiad in Informatics (2023)
- Academic Excellence Scholarship:** Awarded in Semesters 4 and 5
- Competitive Programming:** 54th Place - 2022 ICPC Vietnam National Contest; 56th Place - Code Tour 2024

PROJECTS

Real-time Collaborative Document Editor

Node.js, Socket.IO, Redis, MongoDB

- Reduced MongoDB writes by 43% with WebSocket batching under 600-user Artillery load tests.
- Improved document API throughput by 29–86% using Redis cache and reduced MongoDB P99 query latency by 94% with indexes.
- Validated Redis Pub/Sub horizontal scaling across 3 servers with 2,500 cross-server messages and 14.85ms average latency.

2D Game Development

C++/SDL2, OOP, Game Development

- Developed a C++/SDL2 game with game loop, rendering, input handling, and collision detection.
- Structured gameplay, rendering, and entity logic into modular OOP components for easier debugging and feature extension.
- Earned the highest project score in class for code quality, gameplay completeness, and technical implementation.

SKILLS

Languages: C++ (strong), Python, Java, JavaScript.

Backend: Node.js, REST APIs, authentication, authorization, RBAC, rate limiting, Socket.IO, Redis Pub/Sub, MongoDB.

CS/C++: Data Structures & Algorithms, OOP, debugging, game loop, rendering, collision detection, concurrency basics.

AI/Tools: LLM APIs, semantic search, Docker, Git, Linux, Kaggle, Jupyter Notebook.